STATISTICS

# Statistics Cheat Sheet

The concepts interviewers ask, each explained the way you would say it to a manager.

- **Core concepts in plain words**
- **Which test to use, when**
- **A/B testing step by step**

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# The basics

You will use these every week, not just in interviews.

- **Mean vs median**
  Use median when data is skewed. Average salary in a team with one CEO is misleading. Median is not.
- **Standard deviation**
  How spread out values are. Small SD means values sit close to the average.
- **68 · 95 · 99.7 rule**
  In a normal distribution, that share of values falls within 1, 2 and 3 standard deviations.
- **Outliers (IQR rule)**
  Below Q1 − 1.5×IQR or above Q3 + 1.5×IQR.

# Hypothesis testing

The most asked stats topic. Learn to explain p-value without jargon.

- **Null hypothesis**

  The default claim: nothing changed. The new button did not change conversion.

- **p-value**

  If nothing really changed, how likely is a result this big? Below 0.05 usually means significant.

- **Type I error**

  False alarm. You say the button worked when it did not.

- **Type II error**

  Missed it. The button worked but your test did not detect it. Usually too small a sample.

# Which test to use

Pick by what you are comparing.

- **Two group averages**
  t-test. Average order value, app A vs app B.
- **Three or more averages**
  ANOVA. Average spend across 4 cities.
- **Two categories related?**
  Chi-square. Is gender related to product choice?
- **Two conversion rates**
  Two-proportion z-test. Old checkout 3.1% vs new 3.6%.
- **Two numbers related?**
  Correlation, then regression. Ad spend vs sales.

# A/B testing in 6 steps

Asked at every product company. Say the steps in this order.

- **1. One primary metric**
  For example checkout conversion rate.
- **2. Sample size first**
  Decide significance (0.05), power (80%) and the smallest lift you care about.
- **3. Random split**
  Users, not sessions, so one person sees one version.
- **4. Run full weeks**
  Weekday and weekend behaviour differ. Do not stop early when it looks good.
- **5-6. Test and check guardrails**
  Significant? Then make sure revenue or retention did not drop.